

Pregnancy is a transformative journey that typically lasts around 40 weeks from the first day of the last menstrual period and is divided into three trimesters each bringing unique physical and emotional changes for the mother and developmental milestones for the baby. During the first trimester which spans weeks one through twelve the fertilized egg implants itself into the uterine wall and rapidly begins to develop into an embryo the heart begins to beat around week six the neural tube which forms the brain and spinal cord closes by week seven and by the end of week twelve the baby is referred to as a fetus with all major organs beginning to form limbs fingers and toes becoming visible and facial features starting to take shape. Common symptoms during this period include nausea often called morning sickness though it can occur at any time of day fatigue breast tenderness frequent urination mood swings food aversions or cravings and light spotting known as implantation bleeding. The mother's body produces increased levels of human chorionic gonadotropin hCG progesterone and estrogen which are responsible for many of these early symptoms. Nutrition during the first trimester is critical and a diet rich in folic acid iron calcium vitamin D and omega-3 fatty acids is strongly recommended. Folic acid particularly helps prevent neural tube defects and should ideally be taken even before conception. Foods like leafy greens lentils dairy products eggs nuts seeds and lean proteins are excellent choices. Indian mothers can benefit greatly from traditional foods like methi leaves rajma moong dal ragi and nachni which are rich in iron calcium and folate. It is important to avoid raw or undercooked meat unpasteurized dairy excessive caffeine alcohol and certain herbal supplements during pregnancy. Prenatal vitamins are usually prescribed from the very beginning of pregnancy to supplement dietary intake. The second trimester runs from week thirteen to week twenty-six and is often called the most comfortable phase of pregnancy as nausea and fatigue generally reduce and energy levels improve. During this period the baby grows rapidly and by week twenty the mother may begin to feel fetal movements known as quickening which feels like fluttering or gentle kicks. The baby develops the ability to hear external sounds and responds to light and the senses of taste and touch begin to emerge. The baby's skin is covered in vernix caseosa a white waxy coating that protects the skin in the amniotic fluid and lanugo a fine hair that covers the body. The mother's belly becomes visibly larger and she may experience round ligament pain backaches leg cramps nasal congestion and skin changes including the linea nigra a dark line that appears on the abdomen and darkening of the nipples and skin around the face known as melasma or the mask of pregnancy. Weight gain during the second trimester should be gradual and steady typically around half a kilogram per week. Regular gentle exercise like prenatal yoga walking and swimming is highly beneficial for managing weight improving circulation reducing stress and preparing the body for labour. In Indian settings yoga has deep cultural roots and prenatal yoga specifically tailored for pregnant women has been shown to significantly reduce anxiety improve sleep and increase overall wellbeing. The baby's skeleton which was initially made of cartilage begins to harden into bone a process called ossification and the baby can now make sucking and swallowing movements in preparation for breastfeeding after birth. Anomaly scans typically performed around week eighteen to twenty check the structural development of the baby including the brain heart kidneys spine and limbs and help identify any potential issues early. The second trimester is also a good time to start childbirth education classes explore breastfeeding support groups and begin discussing birth preferences and birth plans with healthcare providers. Third trimester begins at week twenty-seven and extends to week forty when the baby is considered full term. During this period the baby gains significant weight and fat deposits under the skin which help regulate body temperature after birth. The lungs mature and produce surfactant a substance that keeps the air sacs open allowing the baby to breathe after delivery. The baby's brain undergoes rapid development in the third trimester and sleep cycles become more established with periods of activity and rest that the mother can often track. By week thirty-six the baby ideally settles into a head-down position called cephalic presentation in preparation for birth although some babies remain in breech position and may require special management. The mother may experience Braxton Hicks contractions which are mild irregular tightenings of the uterus that are practice contractions

different from true labour contractions which are regular progressively stronger and closer together. Other third trimester symptoms include shortness of breath due to the uterus pushing up against the diaphragm heartburn swollen ankles and feet varicose veins hemorrhoids difficulty sleeping due to the size of the belly and frequent urination as the baby presses on the bladder. Emotional preparation for birth and parenting is as important as physical preparation during this phase and addressing fears about labour pain postpartum recovery and newborn care through counselling classes and peer support is very valuable. Iron deficiency anaemia is very common in Indian pregnant women due to inadequate dietary intake and repeated pregnancies and supplementation along with vitamin C to enhance absorption is a routine part of antenatal care. Gestational diabetes is a condition where blood sugar levels rise during pregnancy and it is particularly prevalent among South Asian women including Malayali women due to genetic predisposition and dietary patterns high in refined carbohydrates. Regular glucose tolerance testing monitoring of blood sugar through diet and exercise and in some cases insulin therapy are part of managing this condition. Hypertensive disorders of pregnancy including gestational hypertension and preeclampsia are serious conditions characterized by high blood pressure protein in urine and in severe cases seizures called eclampsia and require close monitoring and sometimes early delivery. Antenatal care visits which are typically scheduled monthly in the first two trimesters and more frequently in the third trimester involve monitoring the mother's blood pressure weight urine and fundal height measuring the growth of the uterus and assessing fetal heartbeat and position. Ultrasounds are performed at key intervals including the dating scan at six to eight weeks the nuchal translucency scan at eleven to thirteen weeks to assess chromosomal risk the anomaly scan at eighteen to twenty weeks and growth scans in the third trimester. Blood tests during pregnancy typically screen for blood group and Rh factor thyroid function anemia hepatitis B HIV and rubella immunity. The Rh factor is important because if a mother is Rh-negative and the baby is Rh-positive the mother may develop antibodies that could harm future pregnancies and an injection of anti-D immunoglobulin is given to prevent this. Breastfeeding is the optimal form of infant nutrition providing antibodies essential fatty acids and hormones that support immune development brain growth and bonding between mother and baby. The World Health Organization recommends exclusive breastfeeding for the first six months followed by continued breastfeeding alongside complementary foods up to two years or beyond. Colostrum the thick yellowish milk produced in the first few days after birth is extremely rich in antibodies and is often called liquid gold for its protective properties. Common breastfeeding challenges include latching difficulties engorgement mastitis low milk supply and sore nipples and early support from lactation consultants or experienced peers can make a significant difference in breastfeeding success. In Kerala the traditional practice of conducting a ceremony called choroonu or annaprasanam marks the introduction of solid foods typically between five and seven months and rice is traditionally the first food offered often as part of a family ritual. Postpartum recovery involves physical healing of the perineum uterus and abdominal muscles as well as emotional adjustment to new parenthood. Postpartum blues which involve mood swings tearfulness and emotional sensitivity in the first few days after birth are extremely common and are related to the dramatic drop in pregnancy hormones after delivery. Postpartum depression is a more serious and longer-lasting condition affecting around one in five new mothers and is characterized by persistent sadness loss of interest in activities difficulty bonding with the baby anxiety and in severe cases thoughts of self-harm and requires professional support and sometimes medication. The postpartum period traditionally called the forty-day period in many Indian communities is a time when new mothers are encouraged to rest eat nourishing foods and receive help with the baby from family members and traditional practices like oil massages herbal baths and specific dietary preparations like shatavari lehyam or pathyam food are common in Kerala. Newborn care in the first weeks involves learning to recognize hunger cues respond to crying establish safe sleep practices understand normal newborn behaviour including jaundice weight loss and frequent feeding and schedule newborn checkups and vaccinations. The national immunization schedule in India recommends BCG

vaccine at birth followed by hepatitis B oral polio rotavirus pentavalent pneumococcal and measles rubella vaccines at various intervals in the first year of life. Skin-to-skin contact immediately after birth known as kangaroo mother care promotes bonding regulates the newborn's temperature and heart rate supports breastfeeding initiation and has been shown to significantly improve outcomes in premature babies. Mental health support for new mothers in Kerala has historically been limited due to stigma around seeking help but growing awareness and access to telehealth services and community support groups are gradually improving access to care. Emotional wellbeing during pregnancy is deeply connected to physical health and practices like mindfulness meditation journaling creative expression maintaining social connections and gentle physical activity all contribute to better mental health outcomes for mother and baby. Research consistently shows that maternal stress anxiety and depression during pregnancy can affect fetal development including brain development immune function and even the baby's temperament after birth highlighting the importance of comprehensive care that addresses both body and mind throughout the pregnancy journey.